1.5 Addition and Subtraction of Mixed Fractions

Think about the situation



In a joint family of Saravanan, during pongal festival celebration, his grandfather, his father and himself wanted to wear the same colour shirt. The cloth needed for stitching 3 shirts are $2\frac{3}{4}$ m, $2\frac{1}{2}$ m and $1\frac{1}{4}$ m respectively. How many metres of cloth has to be purchased in total?

So the total length of the cloth bought by his father is $2\frac{3}{4}+2\frac{1}{2}+1\frac{1}{4}$. This is solved in the following example.

Example 12 Saravanan's father bought $2\frac{3}{4}$ m, $2\frac{1}{2}$ m and $1\frac{1}{4}$ m of cloth. Find the total length of the cloth bought by him?

Solution Total length of the cloth = $\left(2\frac{3}{4}+2\frac{1}{2}+1\frac{1}{4}\right)$ m

First we add whole numbers: 2 + 2 + 1 = 5 m

Then, add the fractions: $\left(\frac{3}{4}+\frac{1}{2}+\frac{1}{4}\right)=\frac{3}{4}+\frac{2}{4}+\frac{1}{4}=\frac{3+2+1}{4}=\frac{6}{4}=\frac{3\times\cancel{2}}{2\times\cancel{2}}=\frac{3}{2}=1\frac{1}{2}$ m

Therefore, the total length of the cloth bought = $5+1+\frac{1}{2}=6\frac{1}{2}$ m

Example 13 Add: $3\frac{2}{4}+7\frac{2}{5}$

Solution
$$3\frac{2}{4}+7\frac{2}{5}=3+\frac{2}{4}+7+\frac{2}{5}$$
$$=3+7+\left(\frac{2}{4}+\frac{2}{5}\right)$$
$$=10+\left(\frac{10}{20}+\frac{8}{20}\right)$$
$$=10+\frac{18}{20}=10+\frac{9}{10}=10\frac{9}{10}$$

E5HPPS

Think about the situation

One day Anitha's mother bought $5\frac{1}{2}$ litres of milk. She has used only $3\frac{1}{4}$ litres of milk to prepare payasam. How much milk is left? That is $5\frac{1}{2}-3\frac{1}{4}$.

Example 14 In the above situation, find the quantity of milk left over. So, subtract $3\frac{1}{4}$ from $5\frac{1}{2}$

Solution The quantity of milk left over = $5\frac{1}{2}-3\frac{1}{4}$

Here, note that 5 > 3 and $\frac{1}{2}>\frac{1}{4}$

The whole numbers 5 and 3 and the fractional numbers $\frac{1}{2}$ and $\frac{1}{4}$ can be subtracted separately.

Note

This method is applicable only when both integral and fractional parts of minuend is greater than that of the subtrahend.

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